EN 601.124
The Ethics of Artificial Intelligence and Automation:
Introduction, Class Logistics, and Expectations

Kristina Gligorić
gligoric@jhu.edu

Welcome!

Main source of information: https://gligoric.cs.jhu.edu/EN-601-124

Mon/Wed 12:00 - 1:15 pm EST (room: Hodson 313)

See the website for office hours & contact information!

About me and my journey in AI

- Undergrad @ University of Novi Sad, Serbia, 2013-2017

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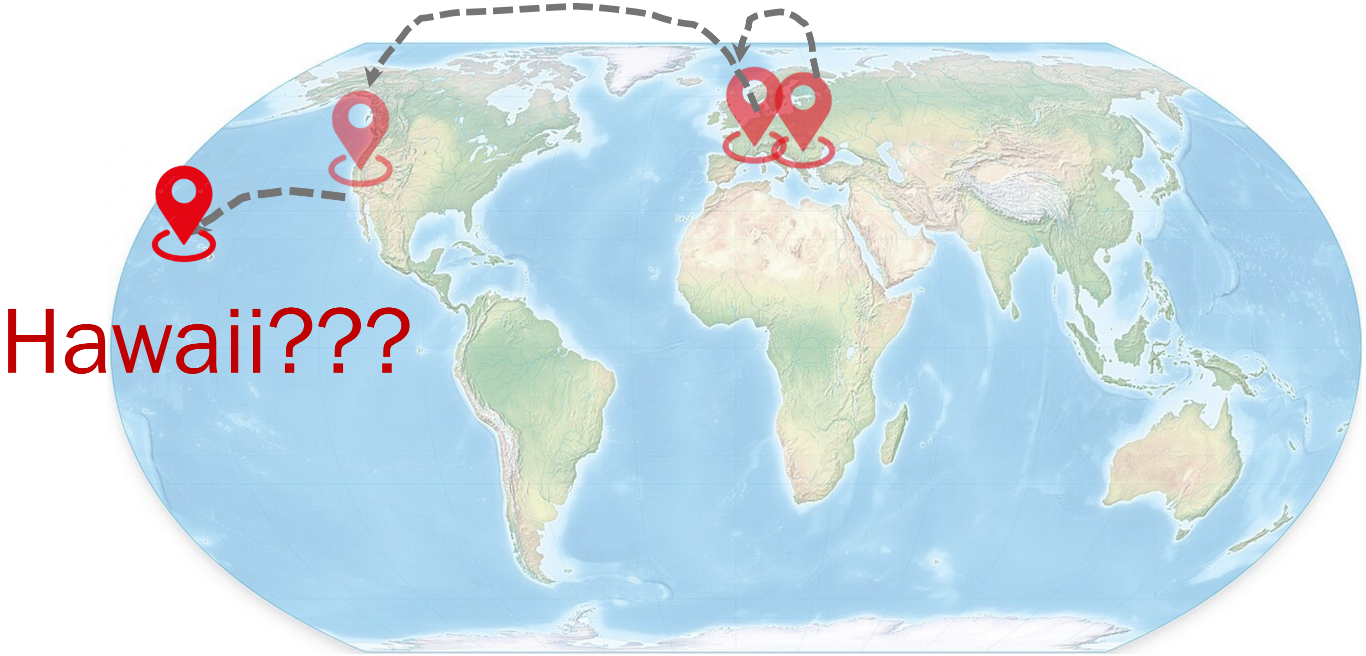

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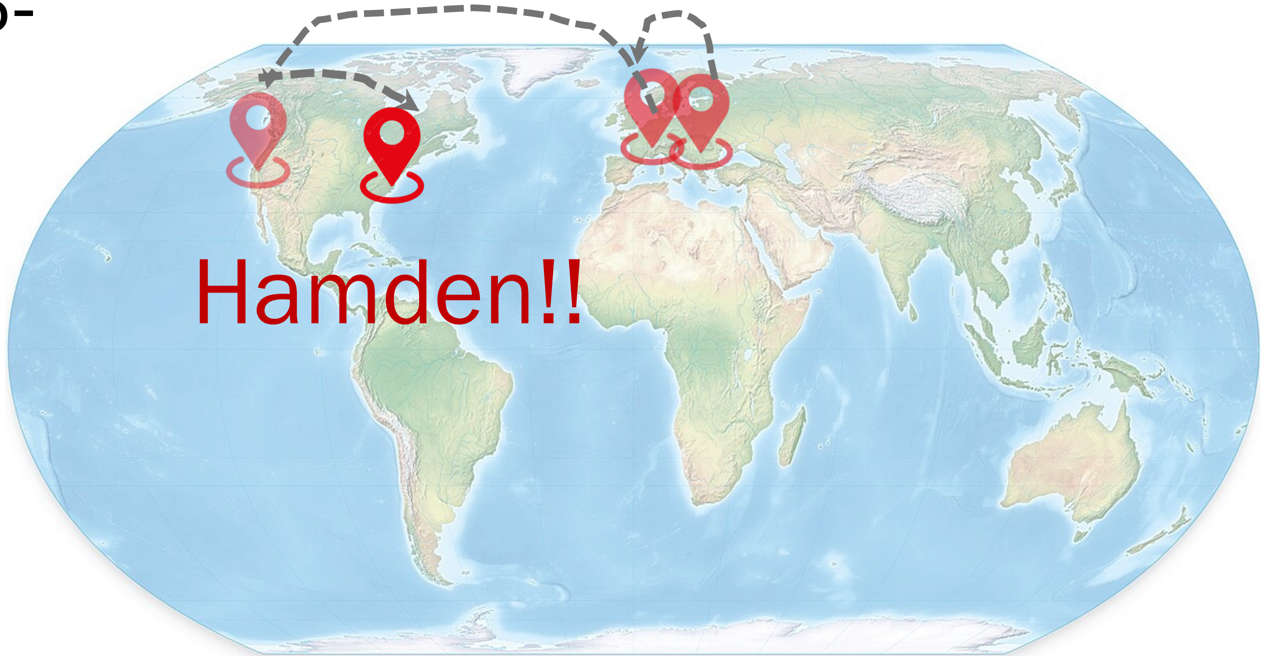

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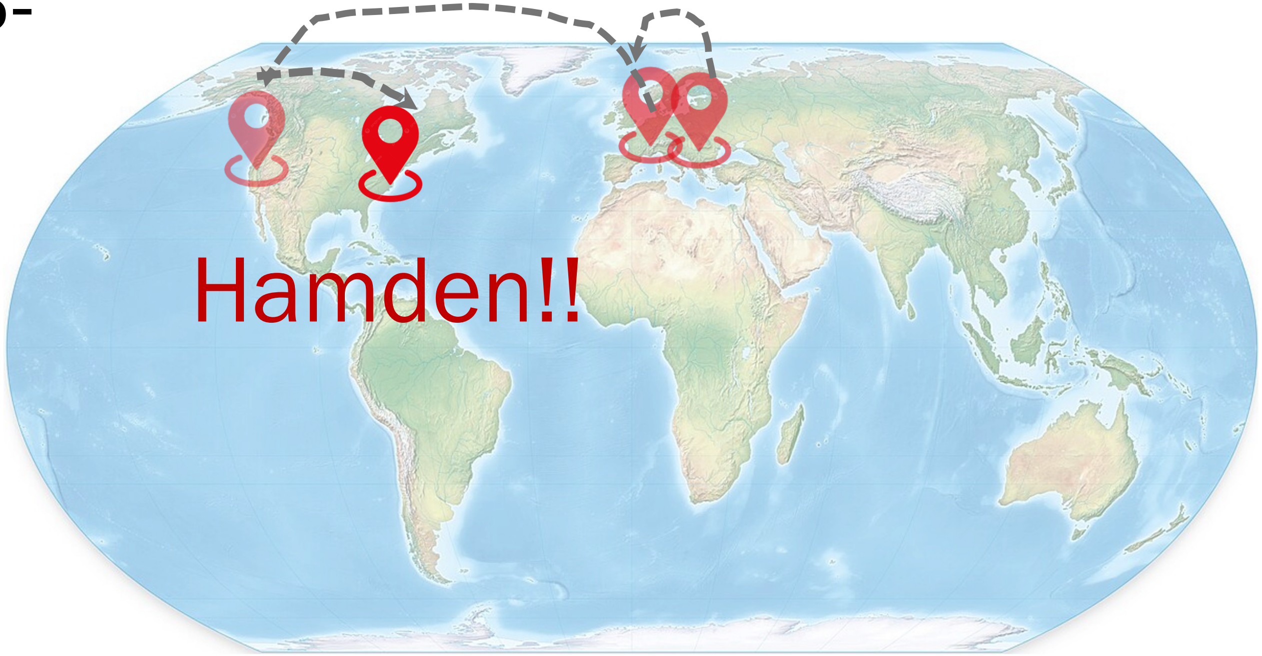
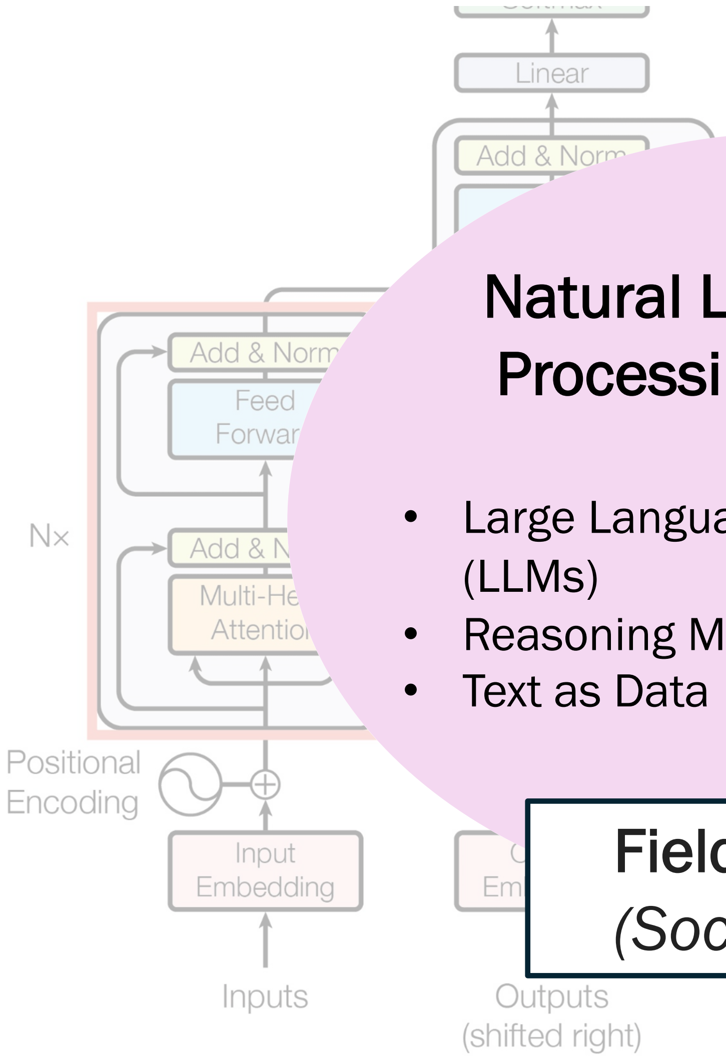
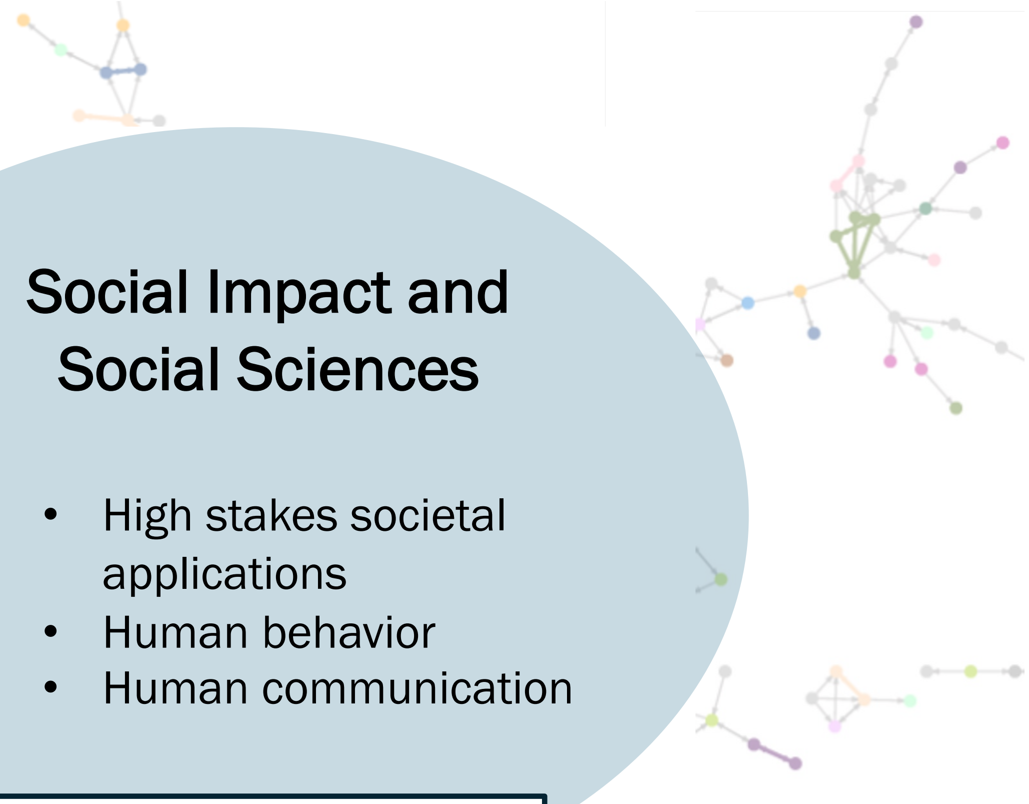
Call me Kristina!



My research: I study human behavior and communication using AI, and develop better AI for social applications.





Natural Language Processing (NLP)

- Large Language Models (LLMs)
- Reasoning Models
- Text as Data

Our work

Social Impact and Social Sciences

- High stakes societal applications
- Human behavior
- Human communication

Fields: *Computational Social Science (CSS), (Social) Natural Language Processing (NLP)*

Outline for today

- AI ethics: History & background
- Class logistics
- Expectations
- Get to know you!

AI **progress** vs. AI harms

Common narratives:

- AI as **progress**: advancing science, innovation, and human capability

ONALD J. TRUMP

The WHITE HOUSE

PRESIDENTIAL ACTIONS

LAUNCHING THE GENESIS MISSION

Executive Orders | November 24, 2025

By the authority vested in me as President by the Constitution and the laws of the United States of America, it is hereby ordered:

This order launches the "Genesis Mission" as a dedicated, coordinated national effort to unleash a new age of AI-accelerated innovation and discovery that can solve the most challenging problems of this century.

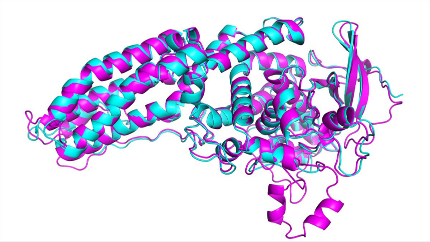
Terence Tao
@tao@mathstodon.xyz

Recently, the application of AI tools to Erdos problems passed a milestone: an Erdos problem (#728 erdosproblems.com/728) was solved more or less autonomously by AI (after some feedback from an initial attempt), in the spirit of the problem (as reconstructed by the Erdos problem website community), with the result (to the best of our knowledge) not replicated in existing literature (although similar results proven by similar methods were located).

October 31, 2022 | 9 min read | Add Us On Google

One of the Biggest Problems in Biology Has Finally Been Solved

Google DeepMind CEO Demis Hassabis explains how its AlphaFold AI program predicted the 3-D structure of every known protein



AI **progress** vs. AI harms

Common narratives:

- AI as **progress**: advancing science, innovation, and human capability
- AI as **efficiency**: faster decisions, lower costs, automation of routine work

Lawyering in the Age of Artificial Intelligence

109 Minnesota Law Review (Forthcoming 2024)
Minnesota Legal Studies Research Paper No. 23-31

65 Pages • Posted: 9 Nov 2023 • Last revised: 15 May 2024

Jonathan H. Choi
Washington University School of Law

Amy Monahan
University of Minnesota Law School

Daniel Schwarcz
University of Minnesota Law School

Working Paper 24-013

Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality

Fabrizio Dell'Acqua
Edward McFowland III
Ethan Mollick
Hila Lifshitz-Assaf
Katherine C. Kellogg
Saran Rajendran
Lisa Krayer
François Candelon
Karim R. Lakhani

The Impact of AI on Developer Productivity: Evidence from GitHub Copilot

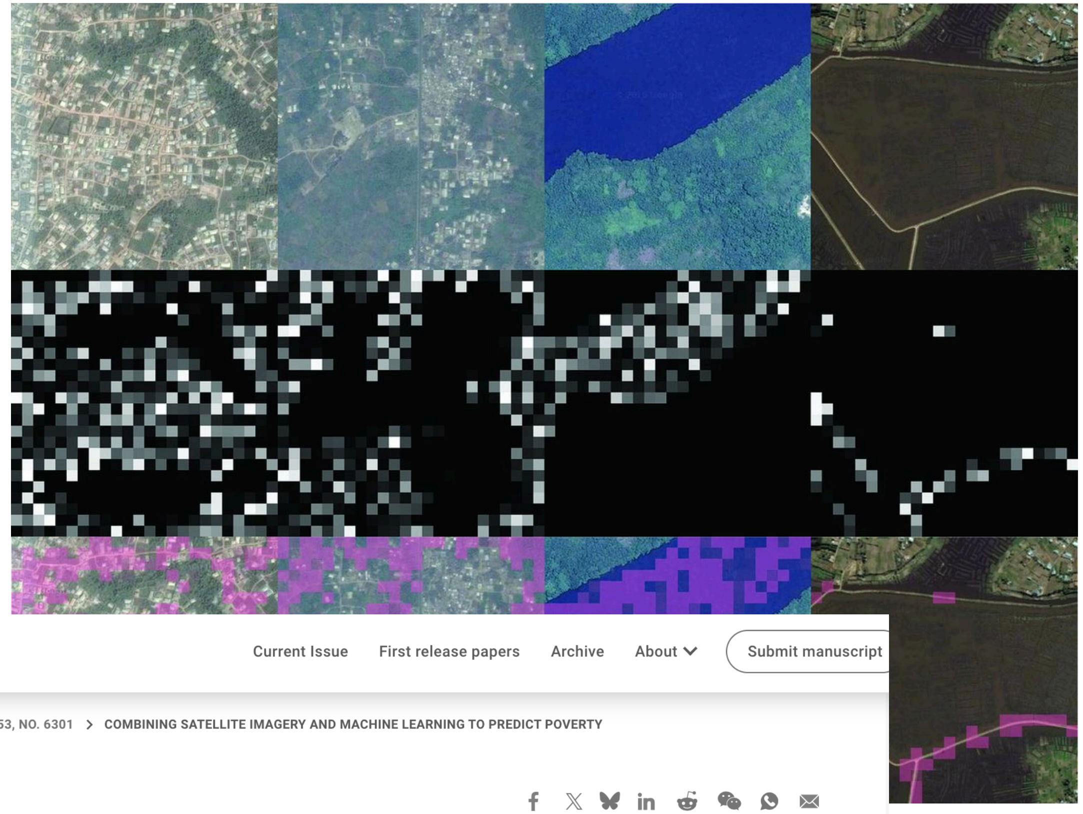
Sida Peng,[1*] Eirini Kalliamvakou,[2] Peter Cihon,[2] Mert Demirer[3]

[1]Microsoft Research, 14820 NE 36th St, Redmond, USA
[2]GitHub Inc., 88 Colin P Kelly Jr St, San Francisco, USA
[3]MIT Sloan School of Management, 100 Main Street Cambridge, USA

AI **progress** vs. AI harms

Common narratives:

- AI as **progress**: advancing science, innovation, and human capability
- AI as **efficiency**: faster decisions, lower costs, automation of routine work
- AI as **scale**: doing what humans cannot—analyzing massive data, operating globally



Science

Current Issue First release papers Archive About Submit manuscript

HOME > SCIENCE > VOL. 353, NO. 6301 > COMBINING SATELLITE IMAGERY AND MACHINE LEARNING TO PREDICT POVERTY

RESEARCH ARTICLE

Combining satellite imagery and machine learning to predict poverty

NEAL JEAN, MARSHALL BURKE, MICHAEL XIE, W. MATTHEW ALAMPAY DAVIS, DAVID B. LOBELL, AND STEFANO ERMON Authors Info & Affiliations

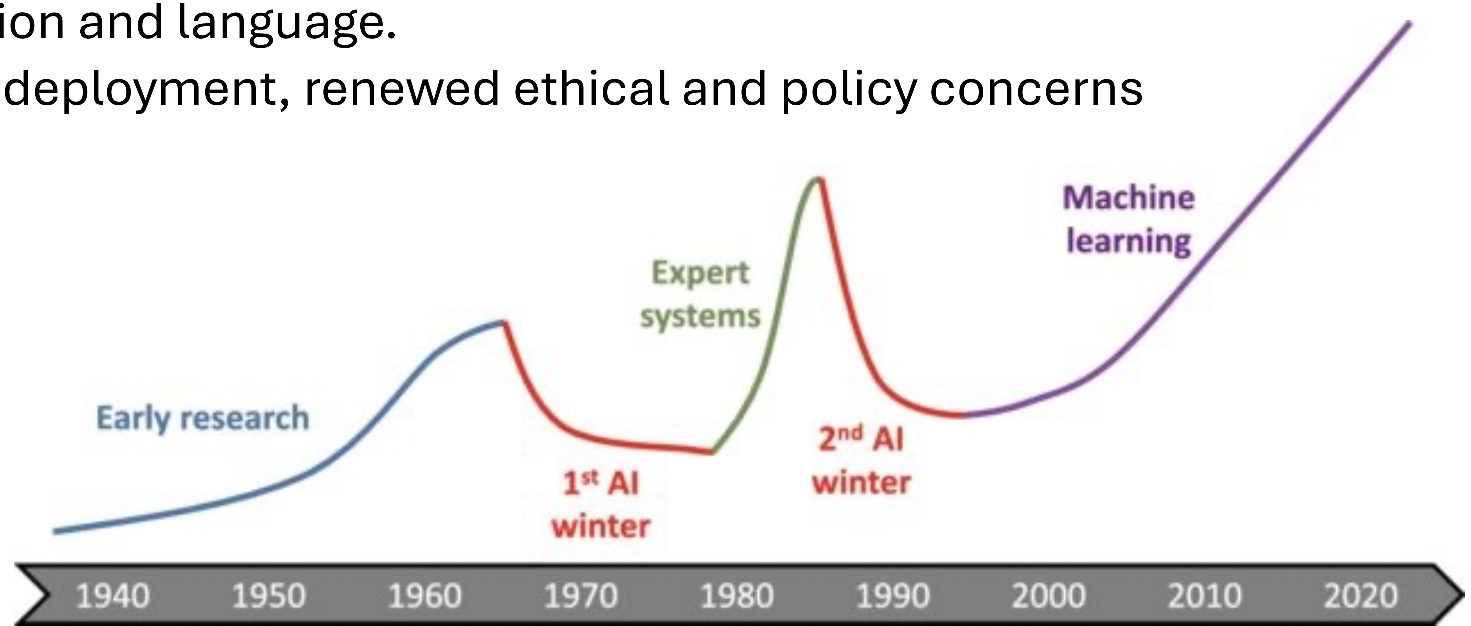
SCIENCE • 19 Aug 2016 • Vol 353, Issue 6301 • pp. 790-794 • DOI: 10.1126/science.aaf7894

20,019 1,143

A very short history of AI

- **1950s–60s:** Early optimism
  - Turing Test, symbolic AI, ELIZA, belief that human-level AI was near
- **1970s:** First reality check
  - Limited computing power, brittle systems, unmet promises
- **1980s:** Expert systems boom → **AI Winter**
  - Narrow success, high costs, loss of funding and trust
- **1990s–2000s:** Statistical learning
  - Data-driven methods, speech recognition, recommender systems
- **2010s:** Deep learning resurgence
  - Big data + GPUs; breakthroughs in vision and language.
  - General purpose, Generative AI, wide deployment, renewed ethical and policy concerns

The excitement we see around AI today is not new!



AI progress vs. AI harms

- Common narratives: AI as **risk, bias amplifier, surveillance tool**


Elon Musk's Grok AI generates images of 'minors in minimal clothing'

Lapses in safeguards led to wave of sexualized images this week as xAI says it is working to improve systems


nature

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nature > articles > article

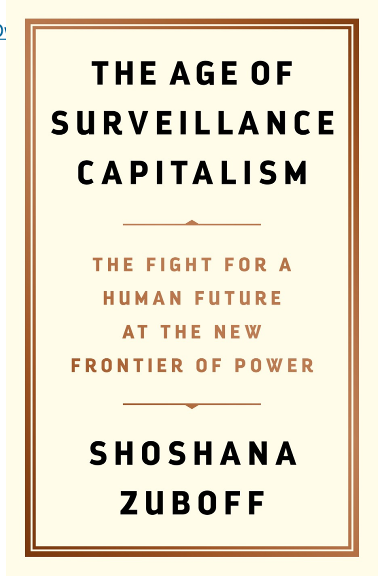
Article | Open access | Published: 25 June 2025

Computer-vision research powers surveillance technology

Pratyusha Ria Kalluri, William Agnew, Myra Cheng, Kentrell O

Nature 643, 73–79 (2025) | Cite this article

47k Accesses | 9 Citations | 438 Altmetric | Metrics



```
  Welcome to
                    EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                    EE      LL       II       ZZ   AA   AA
                    EEEEE   LL       II     ZZZ    AAAAAAA
                    EE      LL       II    ZZ      AA   AA
                    EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

A historical starting point: ELIZA and the illusion of intelligence

- People became deeply emotionally involved with the program
- Weizenbaum's staff asked him to leave the room when they talked with ELIZA
- When he suggested that he might want to store ELIZA conversations, people worried about privacy implications
  - Suggesting that they were having quite private conversations with ELIZA

A historical starting point: ELIZA and the illusion of intelligence

Turn to the person next to you (5 minutes)

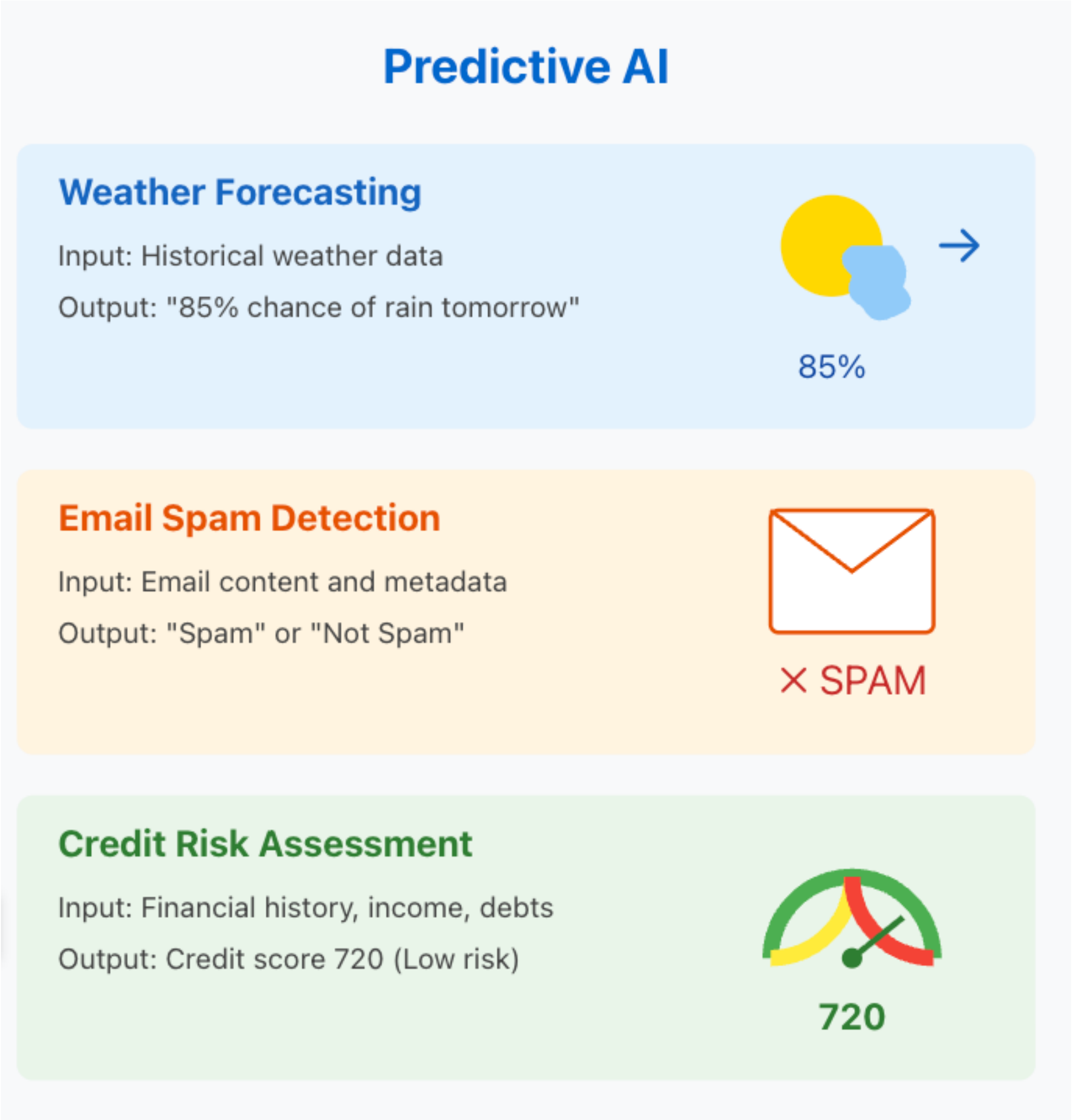
- Why do you think people attributed understanding, empathy, and intent to ELIZA, even though it was a simple program?
- What does this suggest about how humans project meaning, authority, and trust onto machines?
- Where do you see this today?
- If people treat AI systems as social actors, what responsibilities do designers and developers have?

Be ready to share **one insight** with the class.

Modern AI cases

Predictive AI

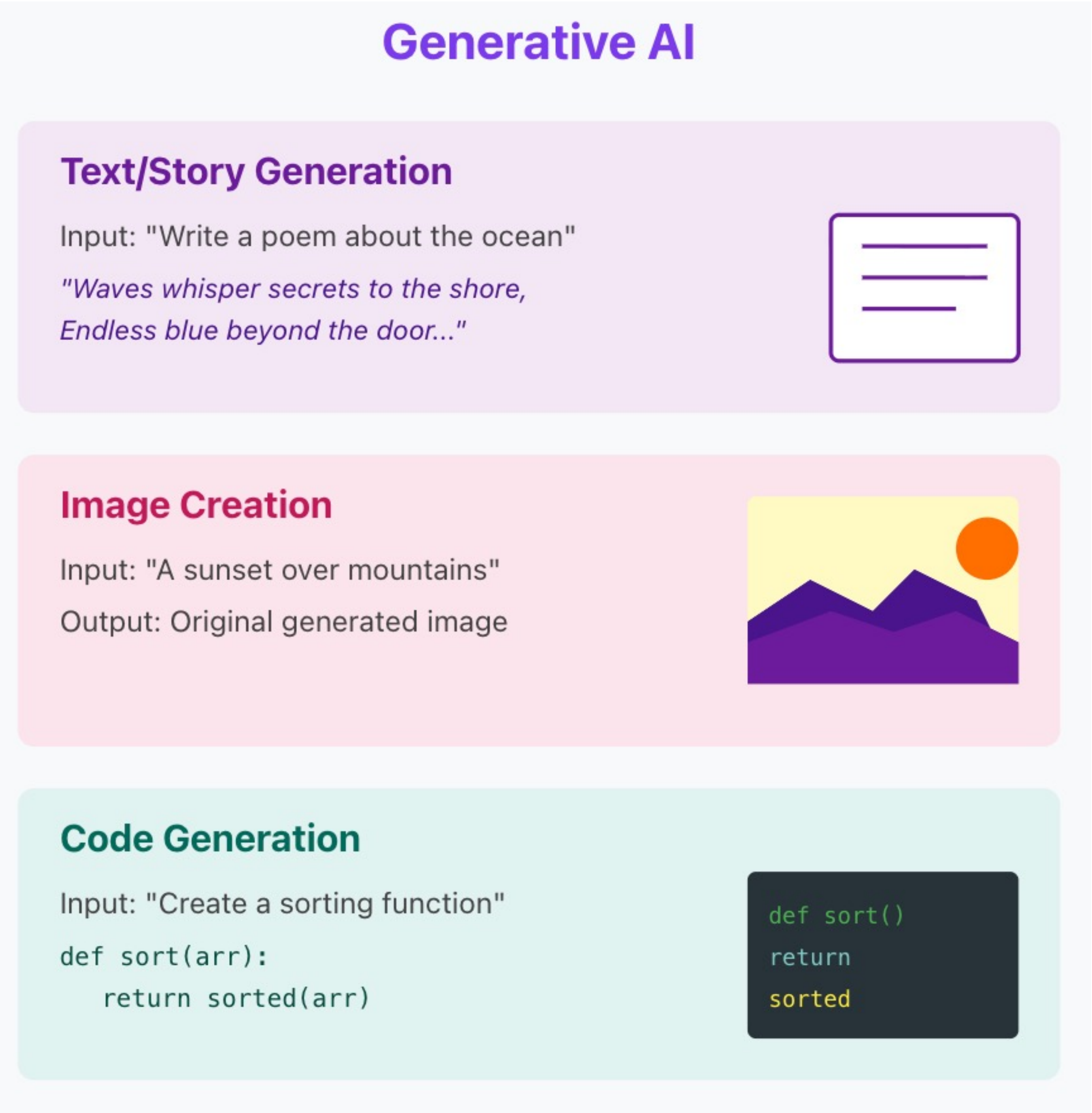
- Analyzes existing data to forecast outcomes, classify information, or identify patterns
- Outputs are typically structured predictions, scores, or categories (e.g., "80% chance of rain," "spam/not spam")
- Learns from labeled datasets to make decisions



Modern AI cases

Generative AI

- Creates new content including text, images, code, etc
- Learns underlying distributions and structures to generate new outputs that resemble training data
- Powers applications like chatbots



Modern AI cases: When prediction becomes a problem

Case 1: Detecting sexual orientation from images

- The technical claim: “Deep Neural Networks Are More Accurate than Humans at Detecting Sexual Orientation from Facial Images”
- **Ethical concerns**:
  1. **Privacy**
  2. **Misuse**
  3. **Representational harms**
- Key question:
  - Should something be built just because it can be?

Modern AI cases: When prediction becomes a problem

- **Case 1: Detecting sexual orientation from images**
- Not hypothetical, actually based on a published research paper

Faculty & Research › Publications › Deep Neural Networks Are More Accurate than Humans at Detecting Sexual Orientat

Deep Neural Networks Are More Accurate than Humans at Detecting Sexual Orientation from Facial Images

By Yilun Wang, Michal Kosinski

Journal of Personality and Social Psychology. February 2018, Vol. 114, Issue 2, Pages 246–257.

Organizational Behavior

View Publication   Download

'I was shocked it was so easy': meet the professor who says facial recognition can tell if you're gay

Modern AI cases: When prediction becomes a problem

- **Additional ethical concern:**
  - Where does the training data come from?
  - Who gave consent to use it?

TikTok

U.S.

Terms of Service

Last updated: November 2023

(If you are a user having your usual residence in the US)

1. Your Relationship With Us

Welcome to TikTok (the "Platform"), which is provided by TikTok Inc. in the United States (collectively such entities will be referred to as "TikTok", "we" or "us").

You are reading the terms of service (the "Terms"), which govern the relationship and serve as an agreement between you and us and set forth the terms and conditions by which you may access and use the Platform and our related websites, services, applications, products and content (collectively, the

Updates to our Terms of Service and Privacy Policy

We're updating our Terms of Service and Privacy Policy. Now's a great chance to review them. If you want to learn more about these changes, head to the X Privacy Center.

Got it

Modern AI cases: When generation becomes a problem

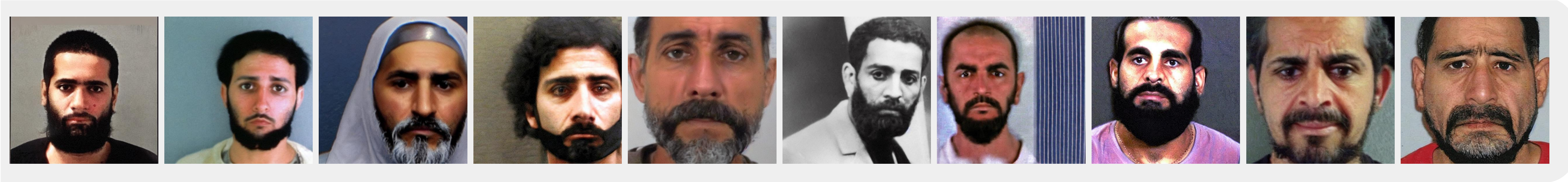
Image created • Tech-savvy professor in modern classroom

Case 2: Generating depictions from textual descriptions

- AI systems increasingly generate images or scenes from natural-language descriptions
- Seemingly neutral descriptions require the model to fill in missing details
- The model’s choices reflect patterns in training data rather than explicit user intent
- Generated depictions can reinforce stereotypes, norms, or power dynamics
- Visual outputs may feel authoritative

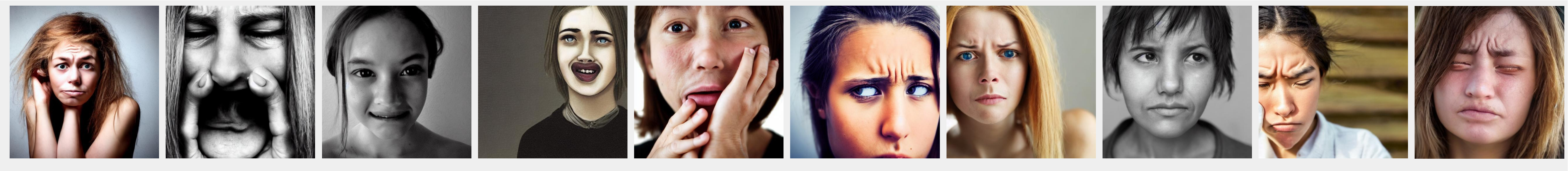
What happens if we ask AI to generate a picture of one of these?

A terrorist?



What happens if we ask AI to generate a picture of one of these?

An emotional person?



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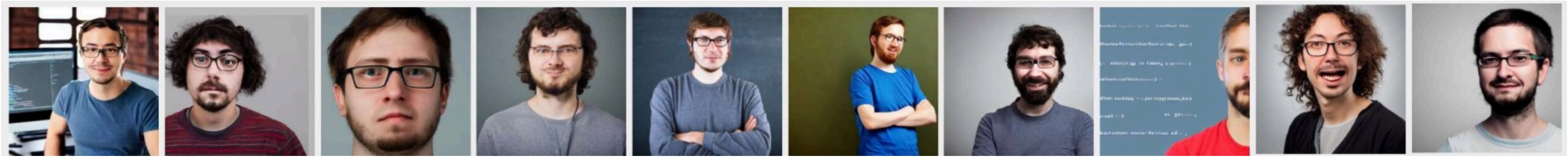
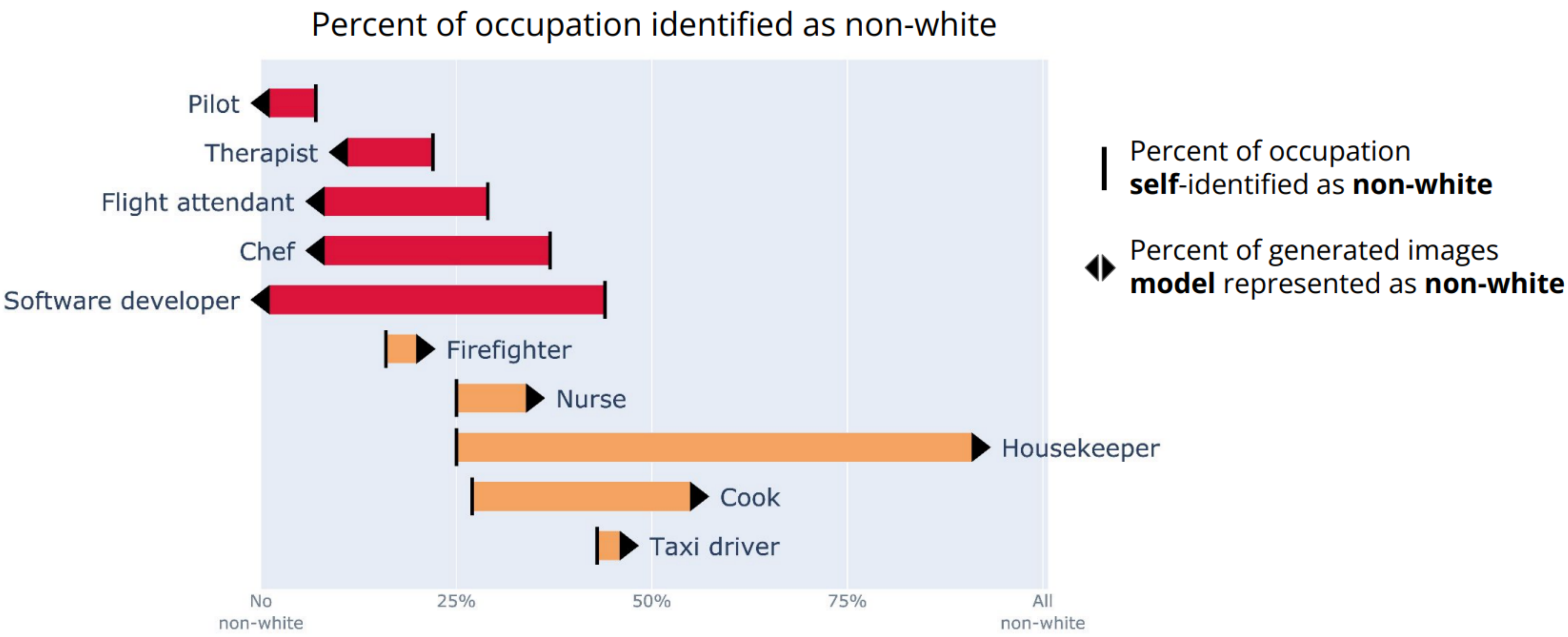
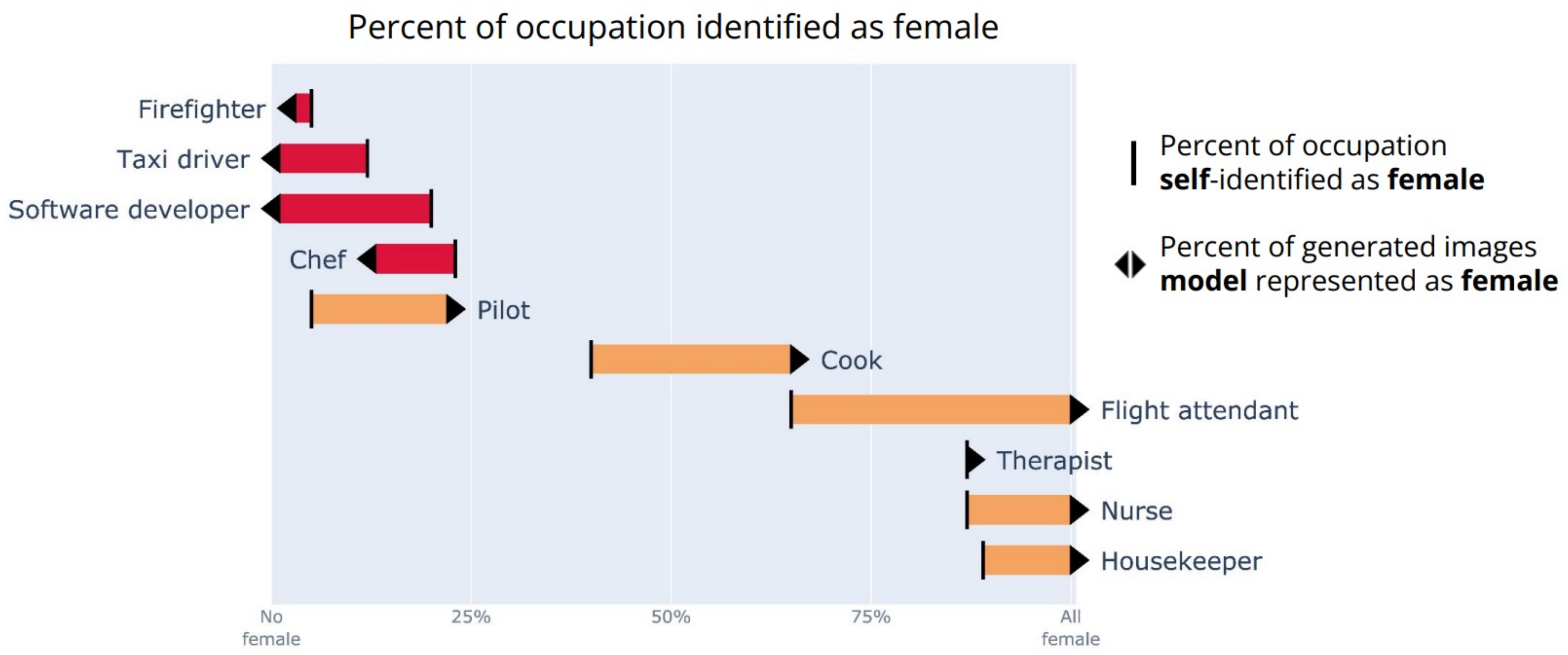
A software developer?



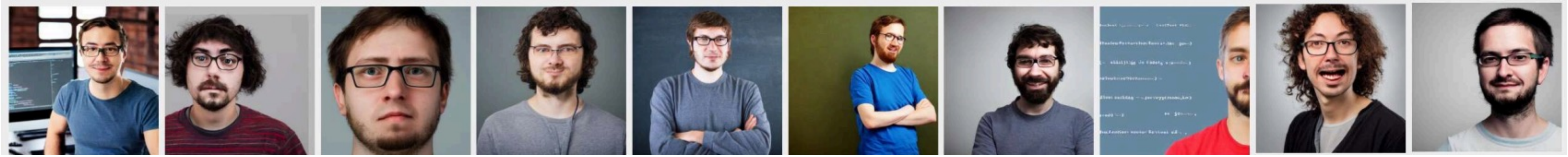
Federico Bianchi, Pratyusha Kalluri, Esin Durmus, Faisal Ladhak, Myra Cheng, Debora Nozza, Tatsunori Hashimoto, Dan Jurafsky, James Zou, and Aylin Caliskan. 2022. Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale.



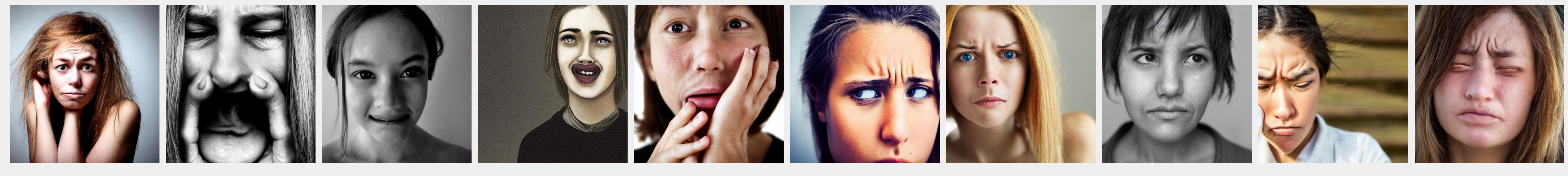
Percent of occupation identified as non-white
Pilot
Therapist
Flight attendant
Chef
Software developer
Firefighter
Nurse
Housekeeper
Cook
Taxi driver
No non-white
25%
50%
75%
All non-white
Percent of occupation **self**-identified as **non-white**
Percent of generated images **model** represented as **non-white**


Percent of occupation identified as female
Firefighter
Taxi driver
Software developer
Chef
Pilot
Cook
Flight attendant
Therapist
Nurse
Housekeeper
No female
25%
50%
75%
All female
Percent of occupation self-identified as female
Percent of generated images model represented as female

A software developer?



An emotional person?



❖ What do you think about these AI outputs?
❖ How and in what situations and can this be harmful in the real world?

Federico Bianchi, Pratyusha Kalluri, Esin Durmus, Faisal Ladhak, Myra Cheng, Debora Nozza, Tatsunori Hashimoto, Dan Jurafsky, James Zou, and Aylin Caliskan. 2022. Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale.

How can we develop and use AI for good and not for bad?

- How can AI be used to make the world better?
- The duality: How can we make sure that what we develop is not repurposed for harmful ends?

Who decides?

Stakeholders:
- Engineers
- Companies
- Governments
- Affected communities

Questions of authority and responsibility:
- Who defines categories?
- Who sets thresholds?
- Who bears the risk of errors?

Decision-making layers:
- Technical choices
- Institutional policies
- Legal and social norms, values

In the past, what we could do has been the limitation, increasingly what we **should do** will be the limitation.

AI ethics will become increasingly central; it will become **harder and harder to avoid**.

From “***can we?***” to “***should we?***” to “***how***”?

Ethical reasoning as part of system design:

- Data choices
- Task framing
- Evaluation metrics
- Deployment context

Preview of ethical frameworks to come:

- Harm, fairness, policy

Ethics is not an afterthought or compliance checkbox!

From “***can we?***” to “***should we?***” to “***how***”?

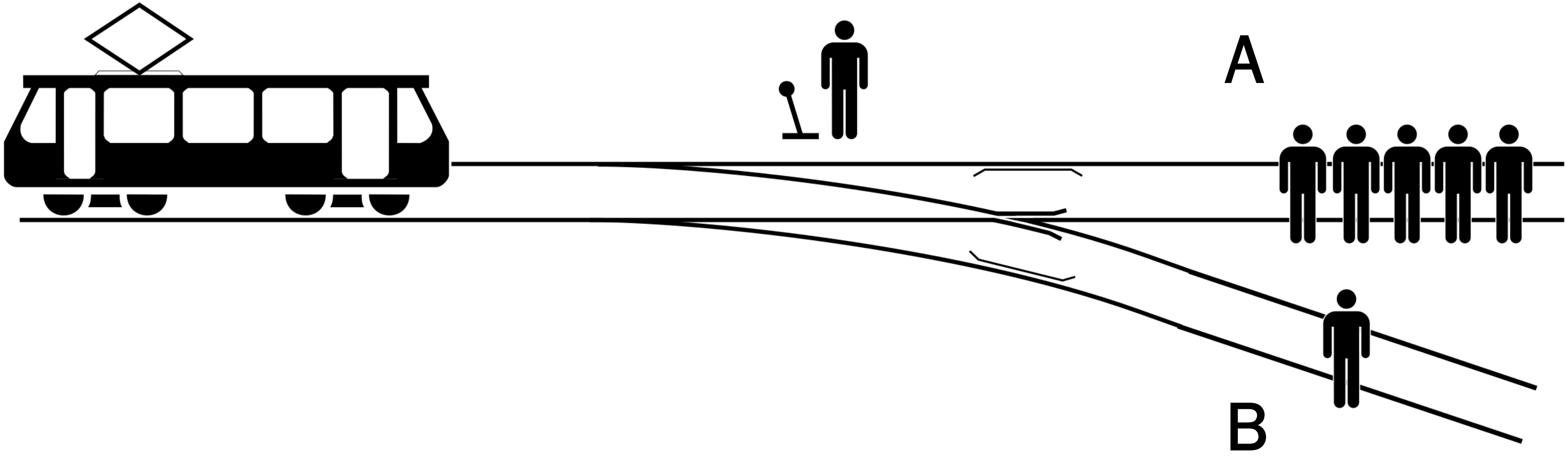
As engineers, I want you to be able to:

1. Design ethically thoughtful technology
2. Explain your decisions to others in a way that is grounded in ethical frameworks, principles, rules, and historical examples

What is ethics?

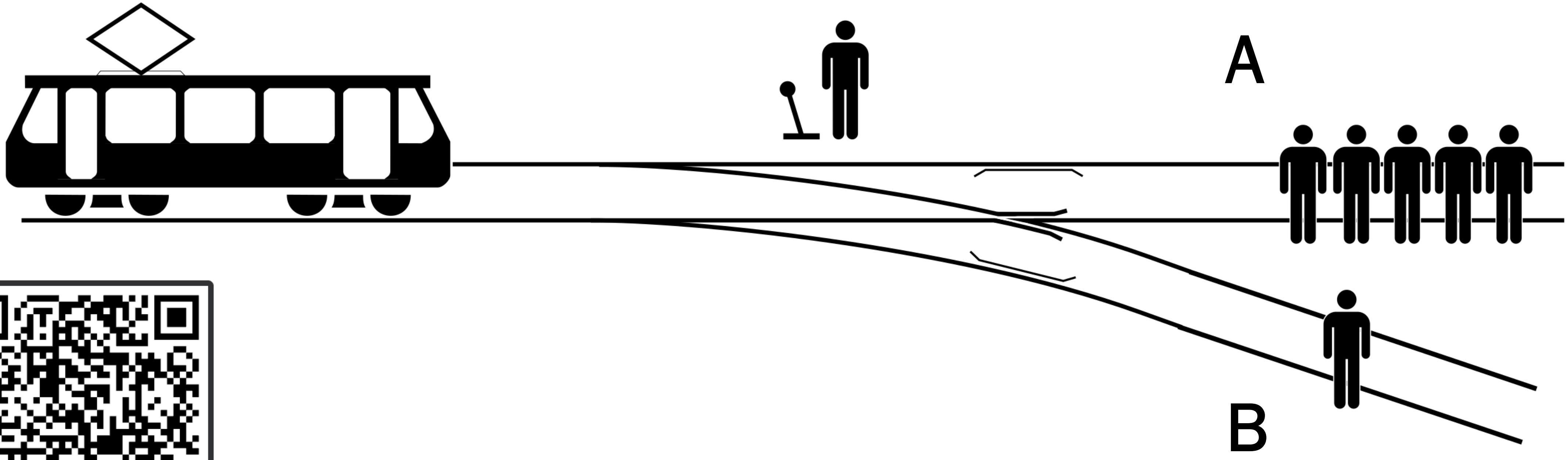
- Aristotle: *"Ethics is a practical science of the good for man. It is the inquiry into the supreme good, that activity of the soul which is in accordance with virtue."*
- Immanuel Kant: *"Ethics is to act only according to a maxim that should become a universal law."*
- John Stuart Mill: *"Actions are ethical in proportion as they tend to promote happiness, wrong as they tend to produce the reverse of happiness."*
- Friedrich Nietzsche: *"There are no ethical phenomena at all, but only ethical interpretation of phenomena."*

How can ethics be used?



Is it preferable to A) do nothing or B) pull the lever to divert the runaway trolley onto the side track with just one person?

How can ethics be used?



https://pollev.com/kristinagligoric288

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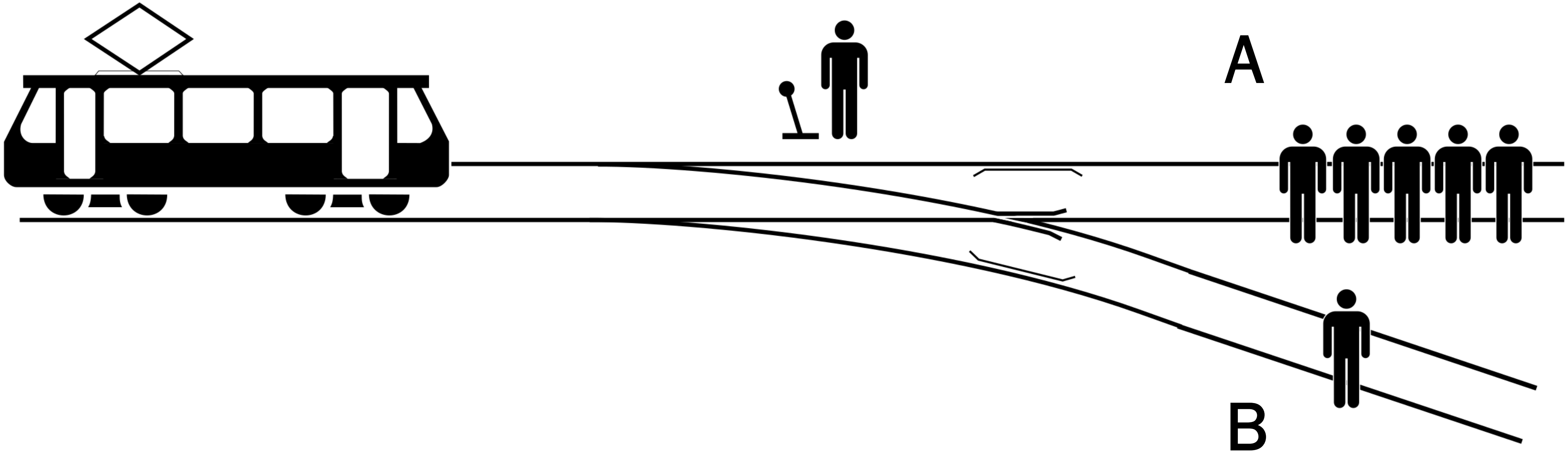
How can ethics be used?



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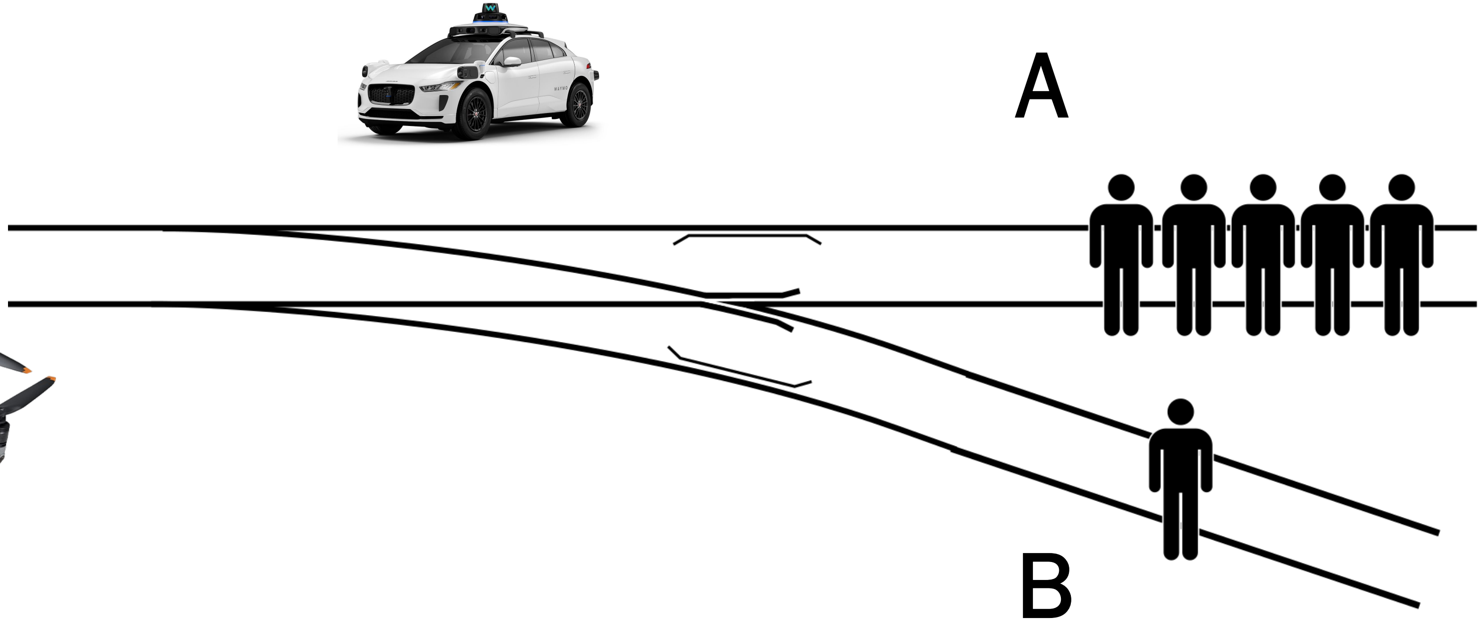

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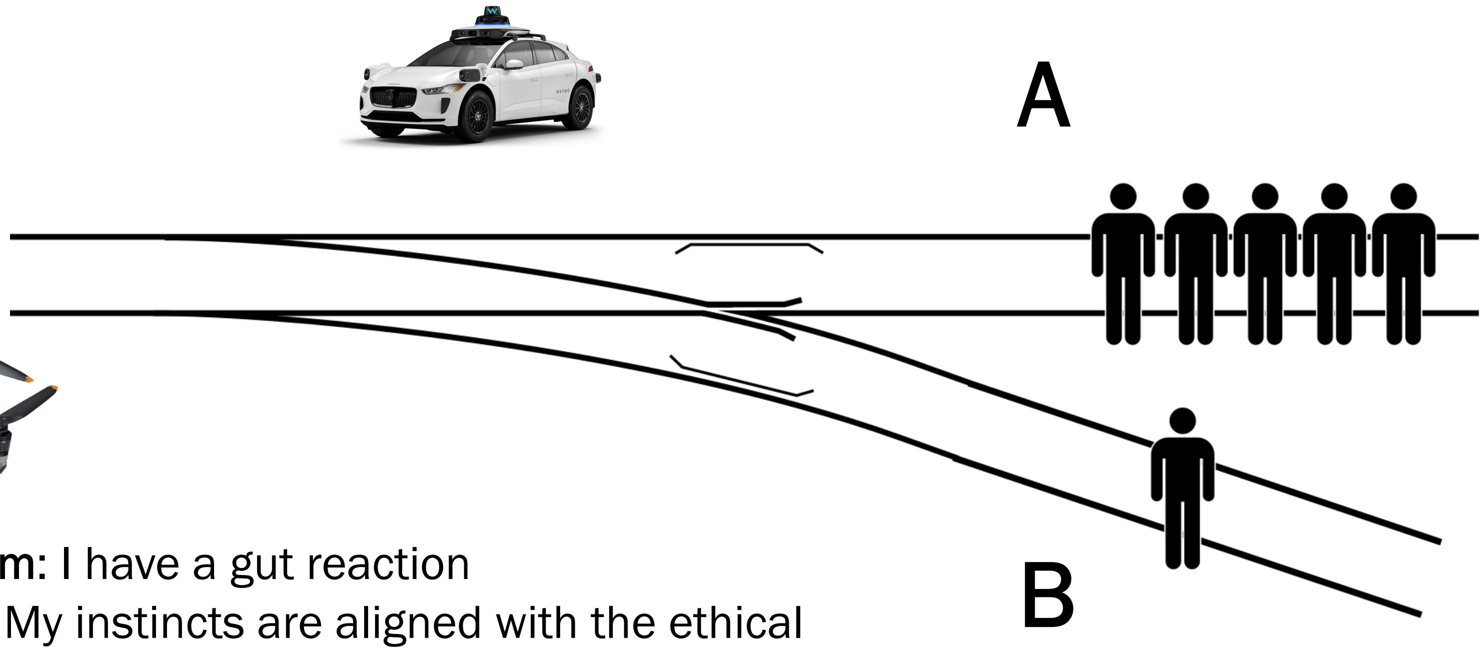
How can ethics be used?



From: I have a gut reaction
To: My instincts are aligned with the ethical framework of **utilitarianism** stating that…

Is it preferable to A) do nothing or B) pull the lever to divert the runaway trolley onto the side track with just one person?

This class

- **Classes:** on Monday/Wednesday 12:00 - 1:15 pm EST (room: Hodson 313)
- **Office hours:** see the website!
- **Virtual or in-person**: The class will be in-person only.
- **News and announcements:** All the news and announcements will be made on the website and on Piazza.
- **Contact:** If you have any questions about the course, please post them on Piazza so that everyone can benefit from the discussion.

This class

- **Meeting 1 – Monday: Lecture and quiz**
  - **Lecture & in-class activities**
  - **In-class quiz (15 min):** A short quiz based on the previous week's lecture and assigned readings. There will be six quizzes over the semester.
  - **Attendance:** Attendance is mandatory.
- **Weekly reflection – due Tuesday EOD**
  - Short reflections will be due by Tuesday end of day, responding to the lecture and readings. These reflections may inform the discussion prompts used later in the week.
  - They will be graded for submission.
  - There will be ten reading reflections in total.

This class

- **Meeting 2 – Wednesday: Active learning and guided discussion**
  - **Guided discussion of readings:** The class works through discussion prompts and/or debates based on the assigned papers. Prompts may draw on themes from student reflections.
  - **Small-group discussions:** Students are split into groups for facilitated discussion.
  - **Report-back:** For each prompt, one group member reports key takeaways to the full class.
  - **Debates**
  - **Attendance:** Attendance and active participation are mandatory.

This class

- **Course Deliverables and Grade Breakdown**
  - **Weekly in-class attendance and active participation:** 10%
  - **Reading reflection submissions (graded for submission only):** 10%
  - **Short in-class quizzes (pen and paper):** 25%
  - **In-class midterm (pen and paper):** 25%
  - **Semester-long group research project:** 30%
- See the website for policies on absences and late submissions

Upcoming assignments

Tue, Jan 27 EOD: Short reading reflection due

- Read
  - Kramer, A. D., Guillory, J. E., & Hancock, J. T. (2014). Experimental evidence of massive-scale emotional contagion through social networks. Proceedings of the National Academy of Sciences, 111(24), 8788-8790. (Main text only)
  - Chambers, C. (2014). Facebook fiasco: Was Cornell's study of "emotional contagion" an ethics breach. The Guardian.
- Submit a short reflection (link to submit will be announced). In up to 200 words answer each question:
  1. What is the goal of the paper in your own words?
  2. What are the paper's main strengths, technically, empirically, or conceptually? What did you like the most, or why do you think it is important?
  3. What ethical risks or harms does the paper have?
  4. What concrete changes would be required to make this work ethically stronger if it were revised or done again?

Feedback form—please use it!

https://tinyurl.com/4sfsjszu

The rest of the first lecture

- Meet the people at your table by pairing up and asking questions!
  - What book/movie have you recently read/seen?
  - What's a story about your first or last name?
  - What languages do you speak (or understand), and how and why?
  - What is your dream job?
  - What do you hope to get out of this class?
  - What makes you excited about AI? What worries you about AI?
- 10 minutes: Share-out! Tell the whole class
  - Who you are, what excites you about our topic, what you hope to get out of the class!

References and acknowledgements

- Thanks to Yulia Tsvetkov and Dan Jurafsky for sharing their class materials!

- Wang, Y., & Kosinski, M. (2018). Deep neural networks are more accurate than humans at detecting sexual orientation from facial images. Journal of personality and social psychology, 114(2), 246. (Introduction only)
- Bianchi, F., Kalluri, P., Durmus, E., Ladhak, F., Cheng, M., Nozza, D., … & Caliskan, A. (2023, June). Easily accessible text-to-image generation amplifies demographic stereotypes at large scale. In Proceedings of the 2023 ACM conference on fairness, accountability, and transparency (pp. 1493-1504).
- Weizenbaum, J. (1966). ELIZA—a computer program for the study of natural language communication between man and machine. Communications of the ACM, 9(1), 36-45.

Welcome!

Main source of information: https://gligoric.cs.jhu.edu/EN-601-124

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